

Robin Carpenter

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RESEARCH EXPERTISE^[3]

Skills and Qualifications

- Data analysis, Python & R (8 yrs.)
- Cleanroom etiquette (*TU Dortmund*)
- Adiabatic Quantum Sim. (*Qiskit*, 3 yrs.)
- Tech. documentation (5 yrs.)
- Optics (1 yr.)
- Linear algebra (2 yrs.)
- Laboratory Optics (1 yr.)
- Presentation and comm. (4 yrs.)

Computing Techniques: **Numerical sim.**, **Monte Carlo sim.**, **quantum and simulated annealing**, **ML**, nonlinear systems, tensor networks

EDUCATION

May 2025

Purdue University – 610 Purdue Mall, West Lafayette IN
BS in *Physics*

Minors in *Planetary Science*, *Spanish for the Professions*,
Astronomy

May 2022

Applications in Data Sciences Certificate

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Data Mine, Purdue University College of Science

- [2021] • **Succeeded** in enhanced background subtraction from dark matter scintillation data (*XENON 1T*, 2021).
- [2022] • **Reported** gaps in Alerce, MARS, TNS for transient photometric measurements
- [2023] • **Contributed** with a team of developers within a Django web environment for the Mara Elephant Project
- [2024] • **Provided** simulated data for the Jung Research Group in quantum computing benchmarking.

[5/2025 – Now]

Quantum Machine Learning Group Member

Recreated 4 methods of measuring non-stabilizerness of quantum states, concerning Second Rényi Entropy and Mana. Currently applying methods to novel QVE ansatz.

[6/2023 – 8/2023]

Data Analysis Intern

Technische Universität Dortmund, Dortmund Germany

Completed project under the ATLAS Phase-II project: developed enhanced angle and edge detection of micron-scale silicon detectors under supervision. Assisted in improving resolution tenfold.

TEACHING EXPERIENCE^[13]

[2023 – Present]

Undergraduate Teaching Assistant,

Department of Physics and Astronomy, Purdue University

• The Data Mine Physics Cohort, Quantum Computing group. **Personally** taught class essentials of quantum information science, organized presentations and group projects, and prepared students for presenting posters at symposiums. Currently focused on *simulating event horizons with differing qubit topologies*.

AWARDS

- *James Griesbach Scholarship*, 2020
- *Shalim and Paula Sargis Memorial Scholarship*, 2021
- *Learning beyond the classroom scholarship*, 2023

SERVICE/LEADERSHIP^[20]

- Purdue Climate Action Committee (*1 yr.*)
- Purdue Bands and Orchestras (*4 yrs.*)
- Purdue Astronomy Club (*1 yr.*)